

Technical Datasheet — V_K HEV Battery Cell

Document Number: VBF-T7R1MIMO-DSH-01

Date: 2026-08-31

Nominal Specifications

Parameter	Value	Unit	Conditions
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Nominal capacity	4.95	Ah	1C, 25°C
Nominal voltage	3.94	V	1C discharge midpoint
Energy density	418.95	Wh/kg	1C, 25°C, dry mass
Volumetric energy density	843.5	Wh/L	1C, 25°C
DC resistance (10% SOC)	0.348	mΩ	1C, 25°C
Power density	269,681	W/kg	V ² /(4·DCR)/mass
Upper cut-off voltage	4.20	V	—
Lower cut-off voltage	2.50	V	—
Cell mass (dry)	43.45	g	formula-caliber
Cell volume	20.62	cm ³	active stack

Rate Performance

Rate	Capacity (Ah)	Retention	Notes
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1C	4.95	100%	baseline
4C charge (no plating)	0.03*	—	anode_potential > 0V; capacity is charge input to cutoff
5C discharge	TBD	—	not simulated

*Note: 4C charge capacity limited by voltage cutoff; key metric is plating-free operation.

Cycling Performance

Condition	Cycles	Capacity Retention	SEI Thickness
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1C/1C, 25°C	100	100% (isothermal model)	—
1C/1C, 45°C	100	— (climb artifact)	536.0 nm

Safety Performance

Test	Condition	Result	Notes
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4C fast charge	45°C, thermal coupled	T_max = 359K (86°C)	No plating
Nail penetration	10W heat, hA=0.5	Not triggered	T_max = 318K (45°C)
Overcharge	+0.5V above cutoff	T_max = 307K (34°C)	No thermal runaway

Operating Conditions

Parameter	Min	Nominal	Max
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Temperature (charge)	0°C	25°C	45°C
Temperature (discharge)	-20°C	25°C	60°C
Charge rate	0.1C	1C	4C
Discharge rate	0.1C	1C	5C